



**FACULDADE DE INFORMÁTICA E ADMINISTRAÇÃO
PAULISTA: ENGENHARIA DE DADOS**

Stream Processing Pipelines

SÃO PAULO
2025

29ABD

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Passo 1: Achar uma base de dados

Usaremos a base de dados de resultados de futebol

https://raw.githubusercontent.com/Git-Mota/base_report/main/results.csv

Passo 2: Rodar no Databricks

Carregar a base

```
11:12 AM (2s) 2
GITHUB_DATA_FILE_URL = "https://raw.githubusercontent.com/heri-macedo/mba-fiap/refs/heads/master/distributed_data_processing/resources/results.csv"
```

```
11:13 AM (<1s) 3
def create_dataframe_schema(column_list: List[str]) -> StructType:
    """
    Creates a StructType object to provide a schema to create a Spark Dataframe.

    Args:
        column_list: List[str] -> List of strings, each one is a column name
    Returns:
        schema: StructType
    """
    fields_list = []
    for column in column_list:
        fields_list.append(
            StructField(
                column, StringType(), True
            )
        )
    schema = StructType(fields_list)
    return schema
```

```
11:13 AM (1s) 4
results_csv_file_content = requests.get(url=GITHUB_DATA_FILE_URL)
content = results_csv_file_content.content
```

✓ 1 minute ago (7s)

```
decoded_content = content.decode('utf-8-sig')
lines = decoded_content.split("\n")
header_line = lines[0]

# Remove empty lines.
data_list = [
    tuple(line.split(","))
    for line in lines[1:]
    if line.strip()
]

header_columns = header_line.split(",")

schema = create_dataframe_schema(column_list=header_columns)

results_df = spark.createDataFrame(data=data_list, schema=schema)

results_df = (
    results_df
    .withColumnRenamed("home_teamName", "home_team")
    .withColumnRenamed("away_teamName", "away_team")
    .withColumn("home_score", col("home_scoreHome").cast("int"))
    .withColumn("away_score", col("away_scoreAway").cast("int"))
    .withColumnRenamed("tournamentName", "tournament")
    .withColumnRenamed("cityCity", "city")
    .withColumnRenamed("countryCountry", "country")
    .drop("neutralTRUE\r")
    .drop("home_scoreHome")
    .drop("away_scoreAway")
)

results_df.cache()
results_df.show(truncate=False)
```

▶ (1) Spark Jobs

▶ results_df: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 6 more fields]

Mostrar que carregou (print, show, count, essas coisas)

```
results_df.show(truncate=False)
```

▶ (1) Spark Jobs

▶ results_df: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 6 more fields]

1873-03-08	England	Scotland	Friendly	London	England	4	2	
1874-03-07	Scotland	England	Friendly	Glasgow	Scotland	2	1	
1875-03-06	England	Scotland	Friendly	London	England	2	2	
1876-03-04	Scotland	England	Friendly	Glasgow	Scotland	3	0	
1876-03-25	Scotland	Wales	Friendly	Glasgow	Scotland	4	0	
1877-03-03	England	Scotland	Friendly	London	England	1	3	
1877-03-05	Wales	Scotland	Friendly	Wrexham	Wales	0	2	
1878-03-02	Scotland	England	Friendly	Glasgow	Scotland	7	2	
1878-03-23	Scotland	Wales	Friendly	Glasgow	Scotland	9	0	
1879-01-18	England	Wales	Friendly	London	England	2	1	
1879-04-05	England	Scotland	Friendly	London	England	5	4	
1879-04-07	Wales	Scotland	Friendly	Wrexham	Wales	0	3	
1880-03-13	Scotland	England	Friendly	Glasgow	Scotland	5	4	
1880-03-15	Wales	England	Friendly	Wrexham	Wales	2	3	
1880-03-27	Scotland	Wales	Friendly	Glasgow	Scotland	5	1	
1881-02-26	England	Wales	Friendly	Blackburn	England	0	1	
1881-03-12	England	Scotland	Friendly	London	England	1	6	
1881-03-14	Wales	Scotland	Friendly	Wrexham	Wales	1	5	
1882-02-18	Northern Ireland	England	Friendly	Belfast	Republic of Ireland	0	13	

only showing top 20 rows

▶ ✓ 2 minutes ago (3s) 6

```
results_df.count()
```

▶ (3) Spark Jobs

40839

Análises:

Análise 1 - Quantas partidas na Copa América o Brasil já ganhou

```

11
df_copa = (
    results_df
    .filter(results_df.tournament == "Copa América")
)

df_copa: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 6 more fields]

12
df_brasil = (
    df_copa
    .filter(df_copa.home_team == "Brazil" | (df_copa.away_team == "Brazil"))
)

df_brasil: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 6 more fields]

13
df_brasil_wins = (
    df_brasil.where(
        ((df_brasil.home_team == "Brazil") & (df_brasil.home_score > df_brasil.away_score))
        | (
            (df_brasil.away_team == "Brazil") & (df_brasil.away_score > df_brasil.home_score)
        )
    )
)

```

103 vitórias:

	^A _C date	^A _C home_team	^A _C away_team	^A _C tournament	^A _C city	^A _C country	¹ ₃ home_score	¹ ₃ away_score
1	1917-10-12	Brazil	Chile	Copa América	Montevideo	Uruguay	5	0
2	1919-05-11	Brazil	Chile	Copa América	Rio de Janeiro	Brazil	6	0
3	1919-05-18	Brazil	Argentina	Copa América	Rio de Janeiro	Brazil	3	1
4	1919-05-29	Brazil	Uruguay	Copa América	Rio de Janeiro	Brazil	1	0
5	1920-09-11	Chile	Brazil	Copa América	Viña del Mar	Chile	0	1
6	1921-10-12	Brazil	Paraguay	Copa América	Buenos Aires	Argentina	3	0
7	1922-10-15	Brazil	Argentina	Copa América	Rio de Janeiro	Brazil	2	0
8	1922-10-22	Brazil	Paraguay	Copa América	Rio de Janeiro	Brazil	3	0
9	1925-12-06	Brazil	Paraguay	Copa América	Buenos Aires	Argentina	5	2
10	1925-12-17	Brazil	Paraguay	Copa América	Buenos Aires	Argentina	3	1
11	1936-12-27	Brazil	Peru	Copa América	Buenos Aires	Argentina	3	2
12	1937-01-03	Brazil	Chile	Copa América	Buenos Aires	Argentina	6	4
13	1937-01-13	Brazil	Paraguay	Copa América	Buenos Aires	Argentina	5	0
14	1937-01-19	Brazil	Uruguay	Copa América	Buenos Aires	Argentina	3	2
15	1942-01-14	Brazil	Chile	Copa América	Montevideo	Uruguay	6	1

103 rows | 0.38s runtime

Análise 2 - Quantas partidas na Copa América o Brasil já perdeu

```

15
df_brasil_loss = (
    df_brasil.where(
        ((df_brasil.home_team == "Brazil") & (df_brasil.home_score < df_brasil.away_score))
        | (
            (df_brasil.away_team == "Brazil") & (df_brasil.away_score < df_brasil.home_score)
        )
    )
)
display(df_brasil_loss)

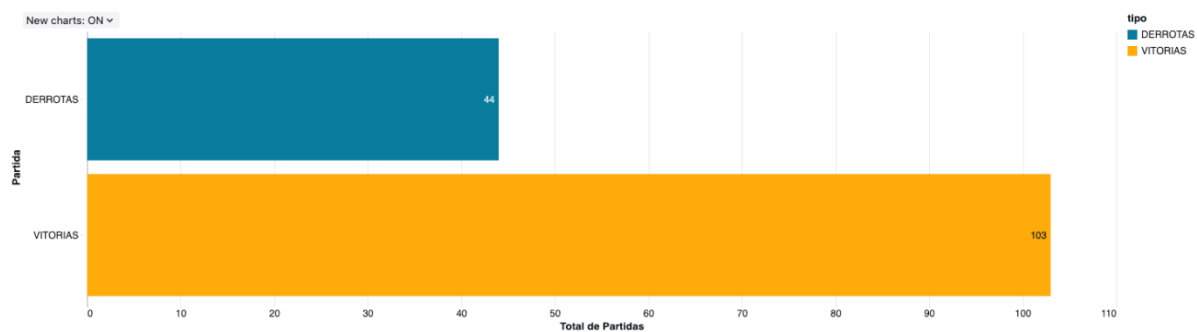
df_brasil_loss: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 6 more fields]

```

44 derrotas:

Table								
	date	home_team	away_team	tournament	city	country	home_score	away_score
1	1916-07-12	Brazil	Uruguay	Copa América	Buenos Aires	Argentina	1	2
2	1917-10-03	Argentina	Brazil	Copa América	Montevideo	Uruguay	4	2
3	1917-10-07	Uruguay	Brazil	Copa América	Montevideo	Uruguay	4	0
4	1920-09-18	Brazil	Uruguay	Copa América	Viña del Mar	Chile	0	6
5	1920-09-25	Argentina	Brazil	Copa América	Viña del Mar	Chile	2	0
6	1921-10-02	Argentina	Brazil	Copa América	Buenos Aires	Argentina	1	0
7	1921-10-23	Brazil	Uruguay	Copa América	Buenos Aires	Argentina	1	2
8	1923-11-11	Brazil	Paraguay	Copa América	Montevideo	Uruguay	0	1
9	1923-11-18	Argentina	Brazil	Copa América	Montevideo	Uruguay	2	1
10	1923-11-25	Uruguay	Brazil	Copa América	Montevideo	Uruguay	2	1
11	1925-12-13	Argentina	Brazil	Copa América	Buenos Aires	Argentina	4	1
12	1937-01-30	Argentina	Brazil	Copa América	Buenos Aires	Argentina	1	0
13	1937-02-01	Argentina	Brazil	Copa América	Buenos Aires	Argentina	2	0
14	1942-01-17	Argentina	Brazil	Copa América	Montevideo	Uruguay	2	1
15	1942-01-24	Uruguay	Brazil	Copa América	Montevideo	Uruguay	1	0

44 rows | 0.41s runtime



Gols Marcados (418 Gols Marcados)

Análise 3 - Quantos gols o Brasil fez

```
Just now (1s) 17

from pyspark.sql import functions as F
df_brasil_total_score = (
    df_brasil
    .where(
        (df_brasil.home_team == "Brazil") | (df_brasil.away_team == "Brazil")
    )
    .withColumn(
        "brazil_goals",
        F.when(df_brasil.home_team == "Brazil", df_brasil.home_score)
        .when(df_brasil.away_team == "Brazil", df_brasil.away_score)
        .otherwise(0)
    )
)

total_brazil_goals = df_brasil_total_score.agg(F.sum("brazil_goals").alias("total_brazil_goals"))
display(total_brazil_goals)
```

▶ (2) Spark Jobs

▶ df_brasil_total_score: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 7 more fields]

▶ total_brazil_goals: pyspark.sql.connect.dataframe.DataFrame = [total_brazil_goals: long]

Table	
1	total_brazil_goals
1	418

Gols tomados (201 Gols sofridos)

Análise 4 - Quantos gols o Brasil tomou

Just now (1s) 19

```

from pyspark.sql import functions as F
df_brasil_total_score_taken = (
    df_brasil
    .where(
        (df_brasil.home_team == "Brazil") | (df_brasil.away_team == "Brazil")
    )
    .withColumn(
        "brazil_goals_taken",
        F.when(df_brasil.home_team == "Brazil", df_brasil.away_score)
        .when(df_brasil.away_team == "Brazil", df_brasil.home_score)
        .otherwise(0)
    )
)

total_brazil_goals_taken = df_brasil_total_score_taken.agg(F.sum("brazil_goals_taken").alias("total_brazil_goals_taken"))
total_brazil_goals = df_brasil_total_score.agg(F.sum("brazil_goals").alias("total_brazil_goals"))
display(total_brazil_goals_taken)

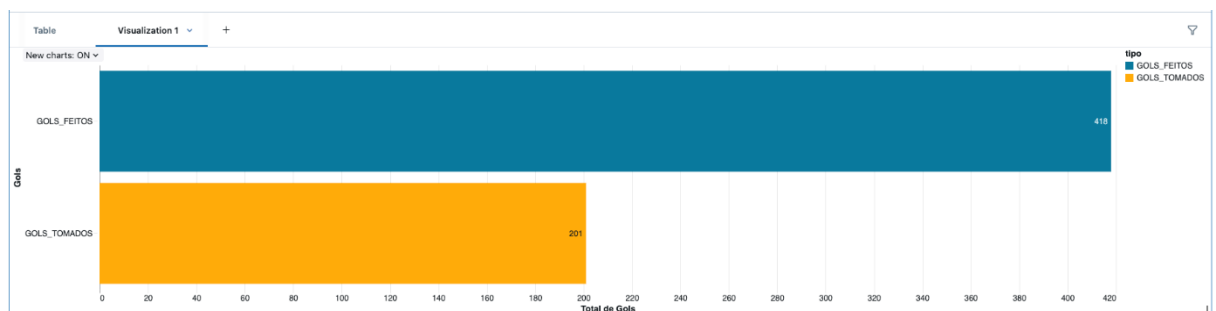
```

▶ (2) Spark Jobs

- df_brasil_total_score_taken: pyspark.sql.connect.dataframe.DataFrame = [date: string, home_team: string ... 7 more fields]
- total_brazil_goals: pyspark.sql.connect.dataframe.DataFrame = [total_brazil_goals: long]
- total_brazil_goals_taken: pyspark.sql.connect.dataframe.DataFrame = [total_brazil_goals_taken: long]

Table +

	total_brazil_goals_taken
1	201





Parte 1: Sobre a base escolhida

Que base é essa?

Base de resultado de futebol de diversas copas entre nações.

Onde acharam?

Conhecemos a base através de disciplinas passadas FIAP.

Quantos registros tem?

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Por que acharam ela interessante?

Conhecemos a base, a estrutura e gostamos do assunto.